

Package ‘treenotypR’

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Description
This R package offers a suite of tools designed for foresters to perform individual tree-level phenotyping. It supports analysis using 3D point cloud data generated from various laser scanning methods, including airborne laser scanning (ALS), unmanned aerial laser scanning (ULS), mobile laser scanning (MLS), and terrestrial laser scanning (TLS). Some functions can also be used with Structure-from-Motion (SfM) photogrammetric point cloud data.
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clip_IT_segments	<i>Clip individual tree segments from a point cloud using crown polygons</i>
------------------	--

Description

This function reads tree crown polygons and then clips individual tree segments based on the provided polygons. Each tree segment is saved as an individual LAS file.

Usage

```
clip_IT_segments(
  las_dir_path,
  crowns_path,
  out_path,
  treeID = "treeID",
  defined_crs = 2193,
  workers = 10
)
```

Arguments

las_dir_path	Character. Path to the directory containing LAS files.
crowns_path	Character. Path to the shapefile containing tree crown polygons. Crowns must have a unique ID, see below.
out_path	Character. Path to the directory where output files will be saved. A new directory will be created here.
treeID	Character. Name of the column in crowns file that uniquely identifies each tree.
defined_crs	Numeric. Coordinate reference system (CRS) for output layers. Default is EPSG:2193.
workers	Integer. Number of workers/logical processors to use for processing. This is used to set up parallel processing. Default is 10.

Details

This function performs the following steps:

- Reads the tree crown polygons from the provided path.
- Reads the LAS files as a catalog.
- Creates an output directory to store the clipped tree segments.
- Clips individual tree segments from the LAS files, saving each tree segment as a separate file.
- Returns the file paths to the clipped LAS files, each representing an individual tree segment.

Value

Clipped individual tree segments will be saved in a newly created directory. A list containing paths to these segments will also be returned.

Examples

```
## Not run:
IT_segments <- clip_IT_segments(
  las_dir_path = "path/to/las/files",
  crowns_path = "path/to/crowns.shp",
  out_path = "path/to/output"
)

## End(Not run)
```

clip_with_buffer	<i>Clip individual tree segments to circular buffer from stem</i>
------------------	---

Description

This function clips individual tree segments from a LAS object using a circular buffer centering at stem.

Arguments

stem_class_las	A LAS object of an individual tree with stem points detected and classified.
buffer	Numeric. The buffer distance from stem (in meters) added to the radius of the clipped segments. Default is 0.5.
min_ht	Numeric. Minimum stem height for segmentation in meters. Points below this height will be excluded. Default is 0.2.
max_ht	Numeric or character. Maximum height for segmentation, either as "zmax" (to use the maximum height in the LAS file) or a specific numeric value. Default is "zmax".
defined_crs	Numeric. Coordinate reference system (CRS) for output layers. Default is EPSG:2193.

Details

This function performs the following steps:

- Filters the LAS data to remove duplicated and NAs.
- Identifies stem points using Hough transformation.
- Loops through height segments to fit circular shapes using RANSAC and clips points within a buffer distance.
- Returns a LAS object with all clipped segments and removes duplicate points.

Value

A LAS object containing the clipped tree segment.

Examples

```
## Not run:
stem_class_las <- detect_stem("path/to/lasfile.laz", stem_only = FALSE)
clipped_las <- clip_with_buffer(stem_class_las, buffer = 0.3)
lidR::plot(clipped_las)

## End(Not run)
```

create_density_raster *Create Density Raster and Calculate Point Density Statistics*

Description

This function creates a density raster for a LAS catalog and calculates point density statistics, providing insights on point distribution.

Usage

```
create_density_raster(las_path, raster_res = 1, defined_crs = 2193)
```

Arguments

las_path	Character. Path to the directory containing LAS files.
raster_res	Numeric. Resolution (in meters) of the density grid. Default is 1 meter.
defined_crs	Integer. EPSG code for the Coordinate Reference System. Default is EPSG:2193.

Details

- Generates a raster of point densities (points per square meter).
- Calculates mean and standard deviation of point density values.

Value

A list containing:

raster	The created density raster.
mean_density	Mean density value across the raster.
sd_density	Standard deviation of the density values.

Examples

```
## Not run:
density_stats <- create_density_raster(
  las_path = "path/to/normalised_tiles",
  out_path = "path/to/output",
  raster_res = 1,
  defined_crs = 2193
)
print(density_stats)
plot(density_stats$density_raster)

## End(Not run)
```

decimate_LAS

Decimate ultra-high-density point clouds to manageable size

Description

This function reduces the density of high-resolution point clouds (e.g., from MLS or TLS) to a manageable size suitable for further processing, such as crown delineation. It performs spatial subsampling using a specified 3D grid resolution.

Usage

```
decimate_LAS(  
  las_dir_path,  
  out_path,  
  grid_res = 0.1,  
  num_points = 1,  
  workers = 10,  
  defined_crs = 2193  
)
```

Arguments

las_dir_path	Character. Path to the directory containing LAS files.
out_path	Character. Path to the directory where decimated LAS files will be saved.
grid_res	Numeric. Grid resolution in meters. A 3D grid of the specified size will be used for decimation. Default is 0.1.
num_points	Integer. Number of points to retain per 3D grid cell in the decimation process. Random selection will be applied. Default is 1.
workers	Integer. Number of workers/logical processors to use for processing. This is used to set up parallel processing. Default is 10.
defined_crs	Numeric. Coordinate reference system (CRS) for the output layers. Default is EPSG:2193.

Details

This function performs the following steps:

- Reads the LAS files from the specified directory as a LAS catalog.
- Checks that the specified num_points is appropriate given the point density of the LAS catalog.
- Applies a decimation process using the specified grid resolution, num_points value and random point selection.
- Saves decimated LAS files in the specified output directory.

Value

A vector of file paths to the decimated LAS files.

Examples

```
## Not run:
decimated_las <- decimate_LAS(
  las_dir_path = "path/to/las_files",
  out_path = "path/to/output"
)

## End(Not run)
```

delineate_crowns	<i>Delineate tree crowns on the canopy height model (CHM)</i>
------------------	---

Description

This function delineates individual tree crowns from a canopy height model (CHM) raster using a watershed segmentation approach. The resulting polygons represent crown boundaries and are matched to the provided stem map.

Arguments

chm_path	Character. Path to the CHM raster file.
stem_map_path	Character. Path to the stem map (point shapefile).
min_ht	Numeric. Minimum height threshold in meters for crown delineation. Any pixels having height below this value will be excluded from the crown delineation process.
smooth	Logical. If TRUE, smooths the edges of delineated crowns. Default is TRUE.
defined_crs	Numeric. Coordinate reference system (CRS) for output layers. Default is EPSG:2193.

Details

This function performs the following steps:

- Reads and processes the CHM raster, applying focal filters to fill gaps and smooth raster).
- Uses watershed segmentation to delineate tree crowns using stem map locations of trees as seed points.
- Label the delineated crowns with matching stem ID.
- Optionally (if specified) smooths the edges of the crown polygons.

Value

A polygon object containing the delineated crown polygons with their matched stem IDs.

Examples

```
## Not run:
crowns <- delineate_crowns(
  chm_path = "path/to/chm.tif",
  stem_map_path = "path/to/stem_map.shp"
)
ggplot2::ggplot(data = crowns) +
  ggplot2::geom_sf() +
  ggplot2::geom_sf_text(ggplot2::aes(label = treeID),
    size = 3, color = "blue"
  ) +
  ggplot2::theme_bw() +
  ggplot2::labs(title = "Crowns with tree IDs")

## End(Not run)
```

detect_stem

Stem detection from LAS individual tree point cloud

Description

This function detects and classifies stem points in an individual tree LAS file, allowing restriction of the height range based on user input. It uses the Hough transformation method to identify and classify the stem cylinder in the point cloud.

Arguments

las_path	Character. The file path to the LAS file.
stem_only	Logical. If TRUE, returns only stem points if FALSE, returns all points classified as Stem and non-Stem. Defaults to TRUE.
min_ht	Numeric. The minimum height threshold in meters for filtering stem points. Points below this height will be excluded.
max_ht	Numeric or character. The maximum height limit in meters for filtering stem points, either as "zmax" to use the file's maximum Z value or a specific numeric height. Default is "zmax".
all_attr	Logical. If TRUE, reads all attributes from the LAS file and keeps them through the processing if FALSE, reads and retains only XYZ coordinates. Defaults to TRUE.
ht_step	Numeric. Height step interval (in meters) used in the Hough transform for detecting stem segments. A smaller value increases accuracy but may increase computation time. Default is 0.1.
max_diameter	Numeric. Maximum allowable stem diameter in meters for classification. Points representing stems larger than this diameter will be excluded. Default is 0.6.
ht_base	Numeric vector of length 2. Specifies the height range (in meters) considered as the base of the tree for stem detection. Default is c(1, 2).
pixel_res	Numeric. Pixel resolution (in meters) for grid-based stem classification. Default is 0.02 meters.

min_den	Numeric. Minimum density of points (per square meter) required for an area to be considered part of the stem. Default is 0.2.
min_numvotes	Integer. Minimum number of votes in the Hough transform required to confirm stem points. Lower values may increase false positives. Default is 3.
defined_crs	Numeric. Coordinate reference system (CRS) for the output LAS object. Default is EPSG:2193.

Details

This function performs stem detection on a LAS file by:

- Reads the LAS file and assigns the specified CRS.
- Checks for missing values in the X, Y, and Z columns.
- Filters points based on the height range defined by min_ht and max_ht.
- Classify stem points using the Hough transformation, restricting results based on parameters like max_diameter, ht_step, and min_numvotes.
- Returns a LAS object containing only the detected stem points meeting these criteria.

Value

A LAS object containing detected stem and non-stem (if specified) points that meet the specified height and classification criteria.

Examples

```
## Not run:
stem_points <- detect_stem(
  las_path = "path/to/your/file.laz",
  stem_only = TRUE
)
lidR::plot(stem_points)

## End(Not run)
```

fixedht_diameter_DF	<i>Calculate diameter at a specified fixed height from a diameter profile</i>
---------------------	---

Description

This function retrieves the estimated diameter of a tree stem at a specified fixed height from a pre-computed diameter profile.

Arguments

diameter_profile	A dataframe containing the diameter profile of a tree with columns 'z' (height) and 'diameter'.
fixed_ht	Numeric. The fixed height (in meters) at which to retrieve the diameter. Default is 1.4 meters.

Details

This function identifies the closest diameter to a specified height by:

- Calculates distance between each z value in diameter_profile and the specified fixed_ht to determine proximity.
- Selects the smallest absolute difference, which represents the diameter measurement closest to fixed_ht.
- Returns a dataframe with the relevant details, including: 'x', 'y', 'z', 'n_points', and 'diameter' (in centimeters).

Value

A dataframe with columns: 'x', 'y', 'z', 'n_points', and 'diameter' (in meters), representing the closest diameter at the specified height.

Examples

```
## Not run:
stem_points <- detect_stem(
  las_path = "path/to/your/file.laz",
  stem_only = TRUE
)
d_profile <- stem_diameter_profile(stem_points)
diameter_at_fixed_ht <- fixedht_diameter_DF(d_profile, fixed_ht = 1.4)
print(diameter_at_fixed_ht)

## End(Not run)
```

fixedht_diameter_LAS	<i>Calculate diameter at a specified fixed height from a LAS object</i>
----------------------	---

Description

This function calculates the diameter of a tree stem at a specified fixed height from an individual tree point cloud.

Arguments

stem_las	A LAS object representing a tree stem point cloud.
fixed_ht	Numeric. The fixed height (in meters) at which to calculate the diameter. Default is 1.4.
seg_ht	Numeric. The vertical interval (in meters) above and below fixed_ht used to define the segment height range. Default is 0.2.
n_points	Integer. Minimum number of points required in the segment for diameter estimation. Default is 25.
quantiles	Numeric vector of length 2. Quantiles for filtering points within the segment based on their distance to the shape center. The values must be between 0 and 1. Default is c(0.1, 0.9).

Details

This function calculates the diameter of a tree stem at a specified fixed height by performing the following steps:

- Extracts a vertical segment of points from the point cloud within a range of $\text{fixed_ht} \pm \text{seg_ht}/2$ meters.
- Filters points within this segment based on their distances from the estimated shape center, using quantiles specified in `quantiles` to remove outliers.
- Fits a shape to the filtered points to estimate the diameter at the specified height.
- Returns a dataframe containing the fitted shape's x and y center coordinates, the specified z height, the count of points `n_points` used, and the calculated diameter.

Value

A tibble with columns: x, y, z, n_points, and diameter (in meters) representing the calculated values at the specified height.

Examples

```
## Not run:
stem_points <- detect_stem(
  las_path = "path/to/las_file.laz",
  stem_only = TRUE
)
diameter_at_fixed_ht <- fixedht_diameter_LAS(
  stem_points,
  fixed_ht = 1.4,
  seg_ht = 0.25
)
print(diameter_at_fixed_ht)

## End(Not run)
```

get_dbh_ht	<i>Detect the nearest internode height where DBH must be measured (For NZ Plantation Forest Inventory: PlotSafe Guideline Replication)</i>
------------	--

Description

This function identifies the nearest internode in a stem profile where DBH must be measured according to the PlotSafe Guidelines. Typically breast height is 1.4 meters for direct measurements. If no direct measurement is possible, this function estimates the nearest height to 1.4 meters where no nodes/swellings are present.

Arguments

node_profile	A dataframe containing the node profile with columns 'z_min', 'z_max', and 'section_type', where 'section_type' denotes "Node" or "Internode".
bh	Numeric. The target height in meters to find the closest internode. Default is 1.4 meters.

min_ht	Numeric. The minimum height in meters to consider for internode detection. Default is 0.2 meters.
max_ht	Numeric. The maximum height in meters to consider for internode detection. Default is 2.9 meters.
tol1	Numeric. Tolerance range (in meters) around the target height for a direct internode measurement. Default is 0.1.
tol2	Numeric. Additional tolerance range (in meters) for finding overlapping internode segments if no direct measurement is possible and split is required. Default is 0.3.

Details

This function detects the nearest internode height to a specified target height, typically used to find the closest internode around breast height (1.4 meters). The function operates as follows:

- Attempts to identify an internode that overlaps with the specified height ('bh') within a small tolerance ('tol1').
- If no direct measurement is available, the function searches for a pair of internode sections, one above and one below the target height, to calculate an estimated height.
- If neither a direct nor split measurement is possible, the function finds the closest available internode below the target height.

Value

A tibble with columns: 'dbh_class', which is the classification of the DBH measurement type ("DirectM", "DirectSS", "DirectLS", or "Split"), list, and 'nearest_int2', an additional internode height or range if a split measurement is needed.

- DirectM: Diameter is measured at breast height (1.4 meters) directly over an internode.
- Direct SS: Diameter is measured directly at a distance of 'tol1' above or below breast height, over an internode (1.3 to 1.5 m).
- Split: To avoid a long welling, diameter measurements are taken at two locations (internodes) over the internodes and then averaged.
- DirectLS: When diameter cannot be measured using the above methods due to the presence of a long node or swollen area, a direct measurement is taken between 0.2 and 2.9 meters over an internode.

Examples

```
## Not run:
stem_points <- detect_stem(
  las_path = "path/to/laz_file.laz",
  stem_only = TRUE
)
d_profile <- stem_diameter_profile(stem_points)
n_profile <- stem_node_profile(d_profile)
dbh_ht <- get_dbh_ht(n_profile,
  bh = 1.4,
  tol1 = 0.1,
  tol2 = 0.3
)
print(dbh_ht)
```

```
## End(Not run)
```

```
get_tree_metrics_ALS    Calculate height and crown metrics from an individual tree point cloud
```

Description

This function calculates height and crown metrics from a point cloud. This is specifically for extracting metrics from airborne laser scanning (ALS) data, including unmanned aerial vehicle laser scanning (ULS). Users can toggle height and crown metric calculations separately and apply a minimum height filter to exclude points below a specified height.

Arguments

las_path	Character. The file path to the LAS file.
height	Logical. If TRUE, calculates height metrics. Default is TRUE.
crown	Logical. If TRUE, calculates crown metrics. Default is TRUE.
min_ht	Numeric. The minimum height (in meters) to filter points. Points with heights below this will be excluded. Default is 0.3.
defined_crs	Integer. Coordinate Reference System (CRS) to assign to the LAS point cloud. Defaults is EPSG:2193.

Value

A dataframe with calculated height metrics (if specified) and crown metrics (if specified).

Examples

```
## Not run:
metrics_ALS <- get_tree_metrics_ALS("path/to/las_file.laz")
print(metrics_ALS)

## End(Not run)
```

```
get_tree_metrics_CHM    Calculate height and crown metrics from CHM
```

Description

This function calculates height metrics for each crown polygon, excluding pixels below a specified minimum height threshold. Optionally, it calculates the crown area for each polygon.

Arguments

chm_path	Character. The file path to the canopy height model (CHM) raster.
crowns_path	Character. The file path to the crown polygons.
height	Logical. If TRUE, calculates height metrics. Default is TRUE.
crown	Logical. If TRUE, calculates crown area. Default is TRUE.
min_ht	Numeric. The minimum height threshold (in meters) for filtering pixels in CHM. Pixels with heights below this will be excluded from metric calculations. Default is 2.
treeID	Character. Name of the column in the crowns data that identifies individual trees. Default is "treeID".
defined_crs	Integer. EPSG code for the Coordinate Reference System to be assigned to both the CHM and crowns. Default is EPSG:2193.

Details

- Height metrics Calculates statistics including max, mean, standard deviation, coefficient of variation, and specified quantiles (5th, 50th, and 95th percentiles) for each crown polygon, excluding pixels below the min_ht threshold.
- Crown metrics Calculates the area of each crown polygon if crown = TRUE.

Value

A dataframe with calculated height metrics (if specified) and crown area (if specified) for each crown polygon.

Examples

```
## Not run:
metrics_CHM <- get_tree_metrics_CHM(
  chm_path = "path/to/chm.tif",
  crowns_path = "path/to/crowns.shp"
)
print(metrics_CHM)

## End(Not run)
```

get_tree_metrics_TLS *Calculate height and crown metrics from an individual tree point cloud*

Description

This function calculates height and crown metrics from a point cloud. This is specifically for extracting metrics from mobile laser scanning (MLS) or terrestrial laser scanning (TLS) data. Users can toggle height and crown metric calculations separately and apply a minimum height filter to exclude points below a specified height.

Arguments

<code>las_path</code>	Character. The file path to the LAS file.
<code>height</code>	Logical. If TRUE, calculates height metrics. Default is TRUE.
<code>crown</code>	Logical. If TRUE, calculates crown metrics. Default is TRUE.
<code>min_ht</code>	Numeric. The minimum height (in meters) to filter points. Points with heights below this will be excluded. Default is 0.3.
<code>defined_crs</code>	Integer. Coordinate Reference System (CRS) to assign to the LAS point cloud. Defaults is EPSG:2193.

Details

This function provides the following metrics:

- Height Metrics: If `height = TRUE`, the function calculates:
 - `maxHt`: Maximum height of the point cloud.
 - `meanHt`: Mean height.
 - `modeHt`: Median height.
 - `stHt`: Standard deviation of heights.
 - `CVHt`: Coefficient of variation of heights.
 - Height percentiles: 5%, 10%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 75%, 80%, 90%, 95%, and 99%.
- Crown Metrics: If `crown = TRUE`, the function calculates:
 - `crownX`, `crownY`: Centroid location of the crown.
 - `concavehull12D_area`: Area of the 2D concave hull.
 - `convexhull12D_area`: Area of the 2D convex hull.
 - `convexhull13D_volume`: Volume of the 3D convex hull.
 - `convexhull13D_area`: Surface area of the 3D convex hull.

These metrics offer insights into canopy spread and crown volume.

Value

A dataframe with calculated height metrics (if specified) and crown metrics (if specified).

Examples

```
## Not run:
metrics_TLS <- get_tree_metrics_TLS("path/to/las_file.laz")
print(metrics_TLS)

## End(Not run)
```

homogenise_LAS

Decimate ultra-high-density point clouds to manageable size

Description

This function reduces the density of high-resolution point clouds (e.g., from MLS or TLS) to a manageable size suitable for further processing, such as crown delineation. It performs spatial subsampling using a specified 3D grid resolution.

Usage

```
homogenise_LAS(  
  las_dir_path,  
  out_path,  
  point_density,  
  grid_res = 0.1,  
  workers = 10,  
  defined_crs = 2193  
)
```

Arguments

las_dir_path	Character. Path to the directory containing LAS files.
out_path	Character. Path to the directory where decimated LAS files will be saved.
point_density	Integer. Point density to be used in homogenisation.
grid_res	Numeric. Grid resolution in meters. A 3D grid of the specified size will be used for decimation. Default is 0.1.
workers	Integer. Number of workers/logical processors to use for processing. This is used to set up parallel processing. Default is 10.
defined_crs	Numeric. Coordinate reference system (CRS) for the output layers. Default is EPSG:2193.

Details

This function performs the following steps:

- Reads the LAS files from the specified directory as a LAS catalog.
- Checks that the specified num_points is appropriate given the point density of the LAS catalog.
- Applies a decimation process using the specified grid resolution, num_points value and random point selection.
- Saves decimated LAS files in the specified output directory.

Value

A vector of file paths to the decimated LAS files.

Examples

```
## Not run:
homogenised_las <- homogenise_LAS(
  las_dir_path = "path/to/las_files",
  out_path = "path/to/output"
)

## End(Not run)
```

segment_diameter	<i>Get diameter from a stem segment point cloud</i>
------------------	---

Description

This function takes a segment of LAS points as input and fits a cylinder to the points, filtering them based on specified distance quantiles and calculating the shape parameters (center coordinates and diameter). It returns 'x', 'y' coordinates at the center and 'diameter' if the fitting is successful.

Arguments

segment	A LAS object representing a segment of stem points.
quantiles	Numeric vector of length 2. Quantiles for point filtering within segments based on distance from the shape center. Values must be between 0 and 1. Default is <code>c(0.1, 0.9)</code> .
n_points	Integer. Minimum number of points required per segment for diameter estimation. Default is 25.

Details

This function performs the following steps:

1. Verifies if the segment contains the minimum required number of points as specified by `n_points`.
2. Attempts to fit a shape to the points in the segment using the RANSAC algorithm via `TreeLS::shapeFit()`.
3. Filters points based on their distances to the initial shape center within the quantile range defined by quantiles.
4. Refits a shape to the filtered points.
5. If the refit is successful, returns the shape parameters (center x, y, and diameter). Otherwise, returns NA values.

Value

A tibble with columns: 'x', 'y', and 'diameter' (in meters) if fitting is successful, or a tibble with 'NA' values if fitting fails.

Examples

```
## Not run:
segment <- readLAS("path/to/las_segment.laz")
diameter <- segment_diameter(segment,
  n_points = 30,
  quantiles = c(0.05, 0.95)
)
print(segment)

## End(Not run)
```

stem_diameter_profile *Calculate stem diameter profile*

Description

This function calculates the diameter profile of a tree stem from an individual tree point cloud.

Arguments

stem_las	A LAS object that has stem points detected and classified.
seg_ht	Numeric. Height of each segment (profile interval) in meters.
min_ht	Numeric. Minimum stem height for segmentation in meters. Points below this height will be excluded. Default is 0.2.
max_ht	Numeric or character. Maximum height for segmentation, either as "zmax" (to use the maximum height in the LAS file) or a specific numeric value. Default is "zmax".
min_diameter	Numeric. Minimum allowable stem diameter in centimeters. Defaults to 0.05.
max_diameter	Numeric. Maximum allowable stem diameter in centimeters. Defaults to 0.6.
quantiles	Numeric vector of length 2. Quantiles for point filtering within each segment, used to exclude outliers in diameter calculation. Default is c(0.1, 0.9).
n_points	Integer. Minimum number of points per segment required for diameter estimation. Default is 25.

Details

This function calculates a diameter profile for a tree stem by performing the following steps:

- Divides the point cloud into vertical segments, each with a height specified by `seg_ht`. Only points within the height range specified by `min_ht` and `max_ht` are used.
- Fits a shape to the filtered points in each segment and calculates the segment diameter. Only segments with at least `n_points` are considered for diameter estimation.
- Returns a data frame where each row represents a segment, including the estimated 'x' and 'y' center coordinates, 'z' height (in meters), point count, and calculated 'diameter' (in meters).

Value

A dataframe with columns: 'ID', 'x', 'y', 'z', 'points', and 'diameter'(in meters), representing the diameter profile of the stem.

Examples

```
## Not run:
stem_points <- detect_stem("path/to/las_file.laz")
d_profile <- stem_diameter_profile(stem_points, seg_ht = 0.25)
print(d_profile)

## End(Not run)
```

stem_node_profile	<i>Detect nodes and internodes from a diameter profile</i>
-------------------	--

Description

This function analyzes the diameter profile of a tree stem to identify nodes (areas of significant diameter increase) and internodes (areas of stability or reduction in diameter). It provides detailed metrics for each identified section, including height range and average diameter.

Arguments

diameter_profile	A dataframe containing the diameter profile of a tree with columns 'z' (height) and 'diameter' in meters.
tol	Numeric. Tolerance threshold for identifying diameter changes in centimeters. Sections with diameter changes larger than this threshold are classified as nodes, while sections with stable diameter are classified as internodes. Default is 1.
base	Numeric. A base value used to inflate the initial segment diameter to assist in node detection. The base will be multiplied by tol and added to the diameter at the min_ht. Default is 3.
split	A vector with a logical value and a character string. The character values must be either "Node" or "Internode". If the logical value is TRUE, only sections of type specified in the string (either "Node" or "Internode") are returned. Default is c(FALSE, "Node").

Details

This function performs the following steps:

- Filters for continuous data, fills or drops missing values, and adjusts segment heights for consistent detection.
- Calculates the diameter difference for each segment and assigns a section type based on the tolerance threshold.
- Summarizes each section by height range, diameter range, average diameter, and section type.
- If specified in `split`, returns only sections of the specified type (either "Node" or "Internode").

Value

A dataframe containing columns: 'section_id', 'z_min', 'z_max', 'z_avg', 'section_length', 'diameter_lower', 'diameter_upper', 'diameter_avg' and 'section_type'. The z values and diameters will be in meters.

Examples

```
## Not run:
stem_points <- detect_stem(las_path = "path/to/las_file.laz")
d_profile <- stem_diameter_profile(stem_points)
n_profile <- stem_node_profile(d_profile)
print(n_profile)

## End(Not run)
```

stem_volume

Calculate cumulative volume of a tree stem

Description

This function calculates the cumulative volume of a tree stem based on its diameter profile, using the formula for the volume of a cylinder for each segment. Missing sectional diameter values can be optionally filled or dropped.

Arguments

diameter_profile	A dataframe containing the diameter profile of a tree with columns 'z' (height) and 'diameter'.
drop.NoData	Logical. If TRUE, rows with missing diameter values are omitted. Default is FALSE.
fill.NoData	A vector with a logical value and a character string. The logical value indicates whether to fill missing diameters, and the string specifies the fill direction ("up", "down", or "updown"). 'up' uses the preceding value, 'down' uses the next value, and 'updown' fills as needed from both directions. Default is c(TRUE, "updown").

Details

This function performs the following steps:

- Checks that 'diameter_profile' contains necessary columns and validates input parameters.
- Identifies the last valid height with data continuity and removes rows above this height.
- Determines the height of each segment based on consecutive 'z' values.
- Fills or drops missing diameter values according to the user input.
- Calculates the volume for each segment and then computes the cumulative volume (in cubic meters) up to each height (in meters).

Value

A dataframe with columns 'vol_min_ht', where the minimum height of the tree for which volume was calculated, 'vol_max_ht', where the maximum height of the tree for which volume was calculated, and 'volume', the cumulative volume (in cubic meters) of the tree stem within the height range.

Examples

```
## Not run:
stem_points <- detect_stem(las_path = "path/to/your/file.laz")
d_profile <- stem_diameter_profile(stem_points)
tree_volume <- stem_volume(d_profile)
print(tree_volume)

## End(Not run)
```

variable_ht_dbh

Estimate diameter at variable height

Description

This function estimates the diameter of a tree stem at variable heights from a diameter profile, using pre-computed internode information.

Arguments

diameter_profile	A dataframe containing the diameter profile of a tree with columns 'z' (height) and 'diameter'.
var_ht	A dataframe generated from 'variable_DBH_ht' with columns specifying breast height ('breast_ht'), tolerance ('splitM_tol'), internode ranges, and DBH classification.

Details

This function retrieves the estimated diameter at a height defined from a specified diameter profile using internode ranges and tolerances:

- Calculates internode segment heights and nearest internode ranges.
- Determines the nearest height segments that overlap with the breast height (bh).
- Based on the 'dbh_class', selects the appropriate diameter measure: "DirectM" retrieves the diameter directly at breast height (1.4m), "DirectSS" finds the diameter closest to the nearest internode, and "Split" averages diameters within a split tolerance range.

Value

A dataframe with columns: 'dbh_class', 'dbh_height', and 'diameter', representing the estimated diameter (in meters) based on height criteria. If the required height measurements or internode data are unavailable, the function returns NA values in the output dataframe.

Examples

```
## Not run:
stem_points <- detect_stem(las_path = "path/to/las_file.laz")
d_profile <- stem_diameter_profile(stem_points)
n_profile <- stem_node_profile(d_profile)
dbh_ht <- get_dbh_ht(n_profile)
dbh <- variable_ht_dbh(d_profile, dbh_ht)
```

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```
print(dbh)
```

```
## End(Not run)
```

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